ENTANGLED MINDS — THE QUANTATHON · QUBITS QREW, NED UNIVERSITY

ECHO — The Signal That Was Never Sent

Investigation Report · Individual submission · Stretch module: 04 — The Erased Measurement

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1 Final Theory and Verdict

> **Physics verdict (high confidence).** ECHO did not demonstrate faster-than-light communication. What the lab observed — two detectors that always agree — is a *correlation*, and my experiments show that agreement alone is not sufficient evidence for a controllable message: a simple pre-agreed classical shared-bit model reproduces the same same-basis statistics as an entangled Bell pair.
>
> **Human verdict (low confidence).** The available evidence does not establish whether Dr. Mahmud made an honest interpretive mistake, whether someone encouraged that interpretation, or whether the experiment was deliberately manipulated. I do not accuse any suspect. The physics verdict is much stronger than any verdict about the people involved.

What I think happened, and who or what was responsible. The best-supported explanation is a reasoning error in the physics: same-basis detector agreement was treated as stronger evidence than it is ("she mistook an echo for a voice"). The evidence identifies that *reasoning flaw* as responsible for the claim; it does not identify any *person* as responsible for the events at the lab.

2 Evidence and Clues

The three clues that shaped my theory

1. **Module 03 Silver — agreement is cheap.** A classical pre-agreed shared bit produced only `00` (1997) and `11` (2003), matching the Bell pair's `11` = 2004, `00` = 1996. Same-basis agreement cannot tell the two apart.
2. **Module 03 Gold — changing basis separates them.** In the X basis the Bell pair still agreed 100.0% (`00` = 1940, `11` = 2060) while the classical mixture agreed only 49.67%. This is the first real crack in ECHO's reasoning — and it is still correlation, not a message.
3. **Module 04 — "changed when observed" is not automatically "hacked".** Inserting a measurement between two Hadamards turned a deterministic `0` into four roughly equal outcomes. A record that differs after observation can be a measurement effect.

Evidence board

Type	Entry
FACT	**M01:** X → `{'1': 1000}`; H → 1020 zeros / 980 ones (P(0)=0.51, P(1)=0.49); H→H → `{'0': 1000}`; 101 → `{'101': 1000}`. **M03:** Bell pair Z-basis agreement 100.0%, mismatch 0.0%; classical shared-bit model also only 00/11; X-basis agreement 100.0% (Bell) vs 49.67% (classical). **M05:** 96-bit stream (32 triples) decoded to `01000101 01000011 01001000 01001111` = **ECHO**. **M04:** H→H = 0; X→H→H = 1; H→X→H = 0; with a measurement inserted, all four joint outcomes appeared (§3).
CLAIM	**Zara:** ECHO sends information faster than light via entanglement. **Vance:** "ECHO was never real." **Sana:** "I only deleted a corrupted log." **Bilal:** "I warned her. Nobody listened." **ARIA:** "No unauthorised command was issued." *Statements, not established facts.*
INFERENCE	Same-basis agreement alone is insufficient evidence for the communication claim. Some post-observation changes can be explained by measurement effects. Over-interpretation of physics is a plausible explanation for Zara's claim.
OPEN	Who configured ARIA's final schedule? What was in Bilal's missing report? Why did Sana delete the log, and what did it contain? Was ECHO deliberately manipulated? What happened to Zara?

3 Experiments and Results

All circuits run on Qiskit Aer (ideal simulator) in Google Colab. Counts below are copied from the saved outputs of the executed notebook. Circuit diagrams are in the supplementary pages.

Module 01 — The Gatekeeper (mandatory)

Purpose: show control of the machine. **Bronze A** X on $|0\rangle$ (prediction: always 1) → `{'1': 1000}`. **Bronze B** H on $|0\rangle$, 2000 shots (prediction ≈ 50/50) → `{'0': 1020, '1': 980}`. **Silver** H→H ($H^2 = I$) → `{'0': 1000}`. **Gold** X on qubits 0 and 2 of $|000\rangle$ → `{'101': 1000}`.

> **What this proves:** the machine reliably prepares deterministic states, superposition, and the required 3-bit pattern. It sets up trust in the tools; it is not a clue about the suspects.

Module 03 — The Two Observers (mandatory; the crux)

Bronze. H on qubit 0, CNOT(0→1) prepares (|00⟩+|11⟩)/√2. Prediction: only 00/11, about half each. Result (4000 shots): `11` = 2004, `00` = 1996; agreement 100.0%, mismatch 0.0%.

Silver. Two classical observers pre-agree on a random shared bit and both output it. Result (4000 samples): `00` = 1997, `11` = 2003 — indistinguishable from the Bell pair in this basis.

Gold. Apply H to both qubits before measurement (X-basis test). Prediction: the Bell pair keeps its agreement; the classical 00/11 mixture drops to about 50%. Bell pair (4000 shots): `00` = 1940, `11` = 2060 → 100.0%. Classical mixture (|00⟩ and |11⟩ each run for 2000 shots, each count halved and summed): `11` = 492, `01` = 519, `00` = 501, `10` = 487 → 49.67% agreement.

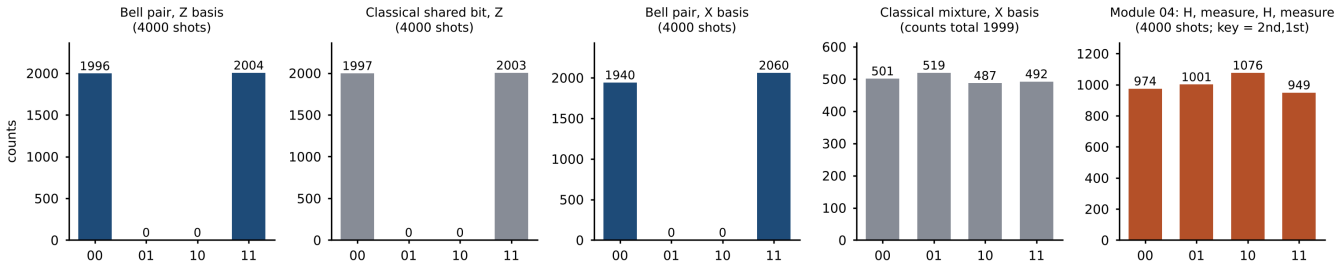


Figure 1. Histograms redrawn from the exact counts printed in the executed notebook (Module 03 Bronze, Silver, Gold; Module 04 Bronze B).

> **What this proves:** same-basis detector agreement alone cannot distinguish an entangled Bell pair from a simple pre-agreed classical correlation. The basis-change test separates the Bell state from *that specific* shared-bit model only.

Module 05 — The Damaged Message (mandatory)

The 96-bit stream (32 triples) was decoded by 3-bit majority vote, giving `01000101010000110100100001001111`, i.e. bytes `01000101`, `01000011`, `01001000`, `01001111` = E, C, H, O → **ECHO**. **Silver:** from `111`, one flip (`101`) decodes correctly to 1, but two flips (`001`) decode wrongly to 0. **Gold:** codewords `000` and `111` have Hamming distance 3, so one flip per triple is corrected; an arbitrary unknown quantum state cannot simply be copied three times (no-cloning theorem), so this classical trick does not protect a qubit directly.

> **What this proves:** majority voting recovered the damaged message as ECHO and reliably corrects one flip per triple, while two flips in one triple can defeat the decoder. It does not show a full quantum error-correcting code.

Module 04 — The Erased Measurement (selected stretch module)

Purpose: compare reversible evolution with evolution interrupted by measurement. **Bronze A** H→H → `{'0': 1000}` (recoverable). **Bronze B** H → measure → H → measure (4000 shots; key = second, first outcome): `11` = 949, `00` = 974, `10` = 1076, `01` = 1001 — all near 25%. The first measurement yields a definite classical bit, so the second H no longer undoes a coherent superposition. **Silver A** prediction X→H→H = X → 1; result `{'1': 1000}`. **Silver B** prediction H→X→H = Z, and Z|0⟩ = |0⟩ → 0; result `{'0': 1000}`. What does the "erasing" is the measurement, not the gates.

Gold preparation / judge prediction practice (not the live Gold task): a helper ran six H/X sequences: H→H → 0; X→X → 0; X→H→H → 1; H→X→H → 0; H alone → `{'0': 507, '1': 493}` (≈50/50); X alone → 1.

> **What this proves:** H and X operations can be reversible, while measurement produces a classical outcome and removes the coherence needed to reverse the earlier evolution in the same way.

4 Interpretation

The case turns on **correlation versus communication**. Module 03 shows that perfect agreement is not itself a message: a pre-agreed shared bit gives the same 00/11 pattern. The changed-basis test shows the Bell pair behaves differently from that classical model, but none of my experiments has one observer choose an input that changes what the other receives. Correlation was observed; signalling was not demonstrated. The "no faster-than-light signalling" conclusion therefore rests on two things: the experiments above (agreement is not evidence of a message) and standard quantum theory's no-signalling property, which this notebook does not itself test — Module 06 would test an instance of it (§6). Module 04 adds that a log which "changed when observed" may reflect measurement, not tampering. Module 05 recovers the word ECHO but, on its own, names no culprit.

What my experiments cannot prove: who configured ARIA's schedule, whether anyone sabotaged ECHO, what Bilal's report or Sana's deleted log contained, or what happened to Zara.

5 Alternative Explanation

The strongest competing theory is **deliberate manipulation or sabotage**: someone configured the schedule, logs or trigger conditions so ECHO appeared more convincing than it was. It stays plausible because Vance had the expertise and a stated motive to see ECHO fail; Sana had privileged log access and admits deleting a log; Bilal reported a discrepancy before the final run and his report is missing; and ARIA ran the schedule — its statement that no *unauthorised* command was issued does not answer who configured the *authorised* instructions. These are leads, not proof, and nothing in my results identifies a saboteur. I prefer over-interpretation of physics because my experiments already reveal a concrete reasoning gap sufficient to undermine the communication claim, whereas sabotage is possible but not required by the evidence.

6 Proposed Next Experiment

Run the Module 06 teleportation circuit from the official hint notebook **(1) with** and **(2) without** the required classical corrections (this is a proposal; Module 06 was not my selected stretch module).

Hypothesis	Expected result
Physics over-interpretation (my theory)	With corrections the receiver recovers the intended state (ideal simulation: near-perfect). Without them the receiver's outcomes should look random and independent of the sent state, so entanglement alone carries no usable message.
Sabotage (alternative)	Module 06 tests the physics, not the people: a correct circuit should give the same results under either hypothesis. Discriminating sabotage would instead need non-quantum evidence, e.g. ARIA's schedule-configuration history and the missing report, if they can be obtained.

7 Limitations

- All experiments used the Qiskit Aer ideal simulator, not real hardware; real devices add noise, decoherence and gate errors.
- The Module 03 Gold test distinguishes the Bell state from the specific pre-agreed shared-bit model only. It is not a loophole-free Bell test and does not rule out every classical hidden-variable model; for instance, a classical strategy pre-agreed for the X basis would reproduce that single-basis result, and only tests with varying measurement bases could address broader models.
- Counts are from a single executed run and the notebook does not fix random seeds, so re-running gives statistically similar but not identical counts (deterministic outcomes such as 100%/0% are unaffected). The classical X-basis counts total 1999 because each 2000-shot run was integer-halved.
- The modules test the physics; they do not solve the human forensic mystery. The word ECHO is not treated as proof of any additional hidden statement beyond what the materials provide.

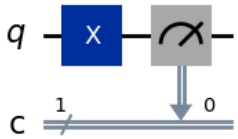
8 References and AI / External Assistance

- Qubits Qrew, NED University. *ECHO — The Signal That Was Never Sent: Participant Manual.* Entangled Minds: The Quantathon, 2026.
- Qubits Qrew, NED University. *ECHO Investigation / Hint Notebook* (official teleportation circuit for Module 06). Entangled Minds: The Quantathon, 2026.
- Qubits Qrew, NED University. *Code of Conduct — Entangled Minds: The Quantathon*, 2026.
- IBM Quantum. Qiskit and Qiskit Aer documentation (circuit construction, simulation, measurement).
- Wootters, W. K. & Zurek, W. H. (1982). A single quantum cannot be cloned. *Nature*, 299, 802–803.
- Nielsen, M. A. & Chuang, I. L. (2010). *Quantum Computation and Quantum Information* (10th anniversary ed.). Cambridge University Press.

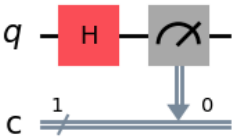
> **AI disclosure.** AI tools were used for code scaffolding, Qiskit syntax assistance, debugging, conceptual explanation, project organization, consistency checking, and documentation drafting. The numerical results reported were generated by executing the submitted Qiskit notebook. AI-generated text or suggestions were not treated as experimental evidence. The participant remains responsible for reviewing, understanding, explaining, and defending the final work.

Supplementary — Circuit diagrams from the executed notebook

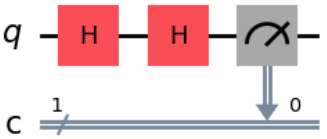
Images below are the circuit diagrams saved in the notebook's outputs (unaltered). Excluded from the 2–3 page main report.



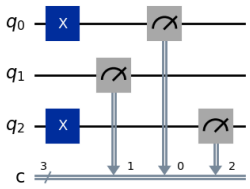
M01 Bronze A: X, measure



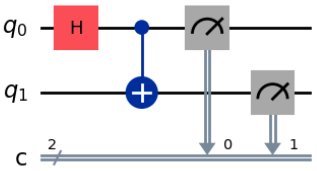
M01 Bronze B: H, measure



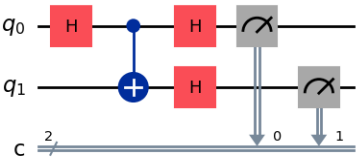
M01 Silver: H→H



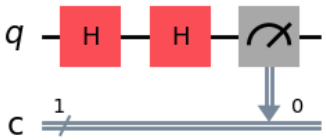
M01 Gold: X on q0, q2 → 101



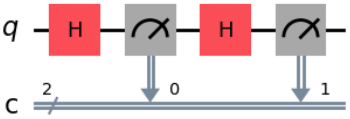
M03 Bronze: Bell pair, Z-basis



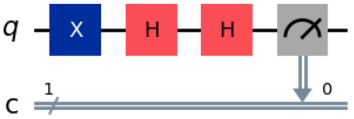
M03 Gold: Bell pair, X-basis



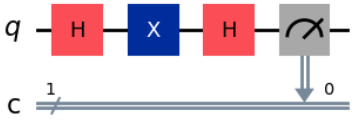
M04 Bronze A: H→H



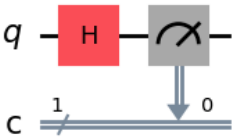
M04 Bronze B: H, measure, H, measure



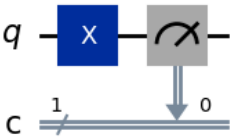
M04 Silver A: X→H→H



M04 Silver B: H→X→H



M04 practice: H alone



M04 practice: X alone

Module 03 Silver (classical shared bit) is a NumPy sampling routine, not a quantum circuit, so it has no diagram. Module 05 is classical decoding code.